

Adhiviraj Pal

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EDUCATION

Galgotias College of Engineering & Technology

B.Tech in Electronics and Communication Engineering

Greater Noida, UP

Aug 2024 – Present (Exp. 2028)

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML, CSS

Frameworks/Libraries: React.js, Next.js, Node.js, Express.js, Django, Tailwind CSS, NumPy, Pandas, Matplotlib, TensorFlow

Tools & Hardware: MongoDB, Git, GitHub, Arduino IDE, ESP32, Model Context Protocol (MCP), REST APIs

PROJECTS

Smart Library System (ESP32 Based)

C++, Arduino IDE, IoT

IoT & Embedded Developer

Greater Noida, UP

- Architected an automated IoT check-in and book allotment workflow using an ESP32 microcontroller, cutting manual desk logging overhead.
- Integrated an optical GM65 QR Code Scanner and SPI-interfaced RFID RC522 reader to execute dual-credential authentication in under 2 seconds.
- Programmed multi-peripheral state machine logic in C++ to handle concurrent events across an I2C OLED display, status LEDs, and buzzer.
- Built persistent entry/exit and inventory tracking routines, reducing human checkout record errors across high student traffic.

Voice-Based Government Scheme Assistant

Python, MCP, External APIs

Python & Backend Developer

Greater Noida, UP

- Developed an AI-powered conversational voice assistant designed to identify and match citizen welfare schemes based on eligibility profiles.
- Implemented Model Context Protocol (MCP) clients in Python to orchestrate dynamic querying and retrieval across external government APIs.
- Built conversational processing pipelines parsing criteria (income, age, demographic constraints) into natural language responses.

ML-Based Laptop Recommendation Platform

Django, React.js, Scikit-Learn

Project Lead & Full-Stack Developer

Microsoft Imagine Cup 2026

- Engineered a recommendation web platform in Django and React.js matching laptop hardware specs with user budget and target workloads.
- Designed an ML recommendation pipeline using feature vectorization to score hardware specs directly against non-technical constraints.
- Led a developer team for Microsoft Imagine Cup 2026, delivering interactive UI components and REST endpoints for instant recommendations.

ACHIEVEMENTS & COMPETITIONS

Microsoft Imagine Cup 2026: Selected as Project Lead for ML-based Laptop Recommendation Platform, leading architecture and full-stack development.

Smart India Hackathon (SIH): Core Team Member; developed an AI-driven Smart Home Automation solution for real-time appliance management.

Algorithmic Problem Solving: Solved 200+ DSA problems across LeetCode and GeeksforGeeks, focusing on graphs, dynamic programming, and trees.

Technical Contests: Participated in multiple software and hardware hackathons, implementing applied ML and embedded prototypes under tight deadlines.